Weiyu Zhou 周伟裕



Material Modeling · CAE / FEA · Test & Reliability R&D

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github.com/weiyuzhou-code

Researcher with doctoral research experience in Solid Mechanics at SUSTech, focusing on viscoelasticity, soft materials and interfacial failure. I connect experimental characterization, constitutive identification and finite element analysis to flexible-display adhesives, camera damping materials and thermal interfaces for chip packaging.

EDUCATION & RESEARCH

Southern University of Science and Technology SUSTech

Solid Mechanics · Combined master–doctoral research / Advisor: Prof. Wei Hong

2026; doctoral degree pending, subject to university approval.

Hefei University of Technology 2015–2019

Hydraulic and Hydropower Engineering · Bachelor's degree / Anhui, China

CORE EXPERTISE

Experimental mechanics
DMA, rheometry, peel/debonding, compression, cyclic and impact tests

Material modeling
Viscoelasticity, hyperelasticity, Prony identification and Mullins softening

Simulation & computing
Abaqus, MATLAB, COMSOL and Python

Images & workflow
Cavity/finger quantification; ImageJ, Git and LaTeX

SELECTED RESEARCH & ENGINEERING PROJECTS

Cavitation & failure of flexible-display adhesives

Designed peel and confined-layer tests and image-processing methods to track cavities and fingers. Combined viscoelastic parameter identification with finite element analysis to connect local damage evolution to mechanical response.

Thermal interface materials for chip packaging

Characterized compression, cyclic loading and rheology. Identified viscoelastic and softening behavior; compared simulations with experiments to assess alignment, size effects and warpage-related softening and recovery.

Curing & dynamic response of camera damping adhesives

Supported UV-curing optimization and developed a high-speed drop-weight impact test platform. Used frequency sweeps, time–temperature superposition and relaxation tests for constitutive modeling and coupon/module validation.

Switchable adhesion of epidermal electrodes

Studied skin–electrode co-deformation and direction-dependent adhesion. Co-first author of the related Advanced Materials publication (2026).

SELECTED PUBLICATIONS

2026 · Advanced Materials · Co-first author
Coordinated Skin-Electrode Deformation for Switchable Adhesion and Injury-Free Detachment of Epidermal Electrodes. 38(37), e73501.

2025 · Small · Co-author
Polyelectrolyte Elastomer-Based Ionotronic Electro-Mechano-Optical Devices. 21(42), e2502225.

2022 · ACS Applied Materials & Interfaces · Second author
Stretchable Heterogeneous Polymer Networks of High Adhesion and Low Hysteresis. 14(43), 49264–49273.